

CONSTRUCTION OS

# PRD - Platform-Wide Integration & Event Architecture

*Implementation-ready Product Requirements Document for Product, Enterprise Architecture, Backend, Frontend, Data, Security, QA, DevOps, SRE and Platform Operations Teams*

**Product purpose**

Define the professional platform-wide contract that allows every ConstructionOS domain to operate as one product without shared writable databases, duplicated business truth, unsafe external exposure or untraceable cross-module side effects.

| Document control       | Value                                                                                                                                |
|------------------------|--------------------------------------------------------------------------------------------------------------------------------------|
| Document               | PRD_Platform_Integration_Event_Architecture                                                                                          |
| Version                | 1.0                                                                                                                                  |
| Status                 | Implementation-ready platform architecture and integration specification                                                             |
| Audience               | Product, architecture, backend, frontend, data engineering, security, QA, DevOps, SRE and technical leadership                       |
| Platform scope         | All internal modules, orchestration services, mobile/offline, client/supplier portals, BIM/Digital Twin, search and analytics        |
| Core architecture      | Domain-owned records + versioned APIs/commands + transactional outbox + event broker + schema registry + permission-safe projections |
| Access model           | Deny-by-default, tenant isolated, active-company aware, project/record scoped, field restricted and externally minimized             |
| Consistency model      | Local ACID transactions within domains; explicit eventual consistency across domains with reconciliation                             |
| Source-of-truth policy | One authoritative owner per business fact; projections and analytics never become competing masters                                  |
| Integration posture    | Self-contained first-party platform core; external adapters remain replaceable and non-authoritative                                 |
| Platform Configuration | Mandatory for event enablement, routing, projection publication, retention, retry, security and operational thresholds               |
| Date                   | 2026-08-06                                                                                                                           |

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# 1. Executive Summary

ConstructionOS is now composed of many mature business domains: Projects, Units, Tasks, Documents, CRM, Contracts, Finance, Procurement, Inventory, Assets, Work Progress, QA/QC, HSE, HR/Payroll, Legal, Calendar, BIM, Portals, Mobile, Reporting, Notifications and Workflow. The final platform risk is no longer missing functionality; it is uncontrolled integration between those domains.

This PRD defines one integration architecture for the complete platform. It prevents shared writable databases, duplicated records, unsafe cross-module joins, hidden dual writes, direct external access to internal entities and module-specific messaging or approval logic. Each domain remains authoritative for its own business facts and communicates through registered, versioned contracts.

## Permanent platform rule

One platform experience. Multiple authoritative domains. Explicit commands, queries, events and projections. No hidden coupling. Complete traceability.

The architecture uses local transactional consistency inside each domain, transactional outboxes for committed facts, at-least-once event delivery, idempotent consumers, version-aware commands, permission-safe projections, centralized workflow and notification orchestration, and deterministic reconciliation. The initial implementation may remain a modular monolith in selected areas, but logical domain boundaries are mandatory from the first release.

# 2. Product Vision and Business Outcomes

- **Unified product behavior** - every surface behaves consistently even though business ownership is distributed.
- **Reliable automation** - cross-module effects retry safely without duplicate invoices, tasks, receipts, notifications or workflow instances.
- **Security** - linked summaries, events, projections and dashboards never expand access to restricted source records.
- **Auditability** - one correlation chain reconstructs the complete path from user command to source changes, events, workflows, notifications and projections.
- **Scalability** - high-volume domains such as Documents, BIM, Notifications, Mobile Sync and Analytics scale independently.
- **Replaceability** - external providers and internal deployment units can evolve without changing business ownership.
- **Development speed** - teams reuse standard integration libraries, schemas and certification tests instead of inventing module-specific contracts.
- **Operational resilience** - temporary outages do not corrupt business records or silently lose required downstream actions.

# 3. Success Metrics and Guardrails

| Metric                   | Initial target                                                                | Guardrail                              |
|--------------------------|-------------------------------------------------------------------------------|----------------------------------------|
| Authoritative ownership  | 100% of governed business entities have one documented owner                  | No shared writable ownership           |
| Direct database coupling | Zero production cross-domain database reads or writes                         | Read models use APIs/projections       |
| Outbox coverage          | 100% of committed cross-domain facts use transactional outbox or approved CDC | No application dual write              |
| Consumer idempotency     | 100% of event consumers pass duplicate-delivery tests                         | At-least-once transport assumed        |
| Contract governance      | 100% of production APIs/events/commands are registered and versioned          | No unregistered production event       |
| Traceability             | >= 99.9% of cross-module operations have trace and correlation IDs            | No confidential payload in logs        |
| Projection freshness     | Configurable SLA; critical operational projections under 60 seconds           | Stale state is visible                 |
| Permission isolation     | Zero cross-tenant, cross-company or unauthorized record exposure              | Permission checks repeated at use time |
| Reconciliation           | All critical flows reconcile automatically with visible exception queues      | No silent mismatch                     |
| Recovery                 | Retry and replay restore recoverable failures without duplicate effects       | Current source state revalidated       |

Guardrails: the platform must not pursue event-driven complexity where a direct authoritative query is safer; it must not create one microservice per screen; and eventual consistency must never be presented to users as immediate finality.

## 4. Scope, Non-Goals and Assumptions

### In scope

- Complete domain ownership and integration boundary model.
- Canonical IDs, references, commands, queries, events and projection envelopes.
- Event broker, schema registry, outbox, consumer, retry and dead-letter standards.
- Workflow, Notifications, Documents, Mobile, Portals, BIM and Analytics integration contracts.
- Cross-module security, audit, trace, observability and reconciliation.
- Reference end-to-end flows and implementation sequencing.
- Contract testing, integration certification and acceptance criteria.

### Non-goals

- Defining the complete database schema for every module; that is the next consolidated domain-model deliverable.
- Replacing module PRDs or changing confirmed source-of-truth decisions.
- Requiring immediate decomposition into dozens of independently deployed microservices.
- Using an enterprise service bus to contain business logic or transform ownership.
- Allowing unrestricted third-party event publication or arbitrary message sending.
- Providing vendor-specific cloud deployment instructions in this PRD.

### Assumptions

- Every domain supports tenant and active-company context.
- Every controlled record has an immutable stable ID and optimistic version.
- Authorization can evaluate company, project, record and field scope.
- Documents, Workflow, Notifications and Audit are platform services available to all modules.
- All timestamps are stored in UTC and presented in configured time zones.

## 5. Architectural Principles

| ID      | Principle                     | Required behavior                                                                              |
|---------|-------------------------------|------------------------------------------------------------------------------------------------|
| INT-P01 | Single authority              | Every business fact has one authoritative domain and one lifecycle owner.                      |
| INT-P02 | No database coupling          | Domains communicate only through approved APIs, commands, events and projections.              |
| INT-P03 | Local consistency             | Business transactions and their outbox records commit atomically in the owning domain.         |
| INT-P04 | Explicit eventual consistency | Cross-domain state is synchronized asynchronously with visible freshness and reconciliation.   |
| INT-P05 | Permission at every boundary  | Authorization applies before queries, projections, delivery, exports and source-record opens.  |
| INT-P06 | Version everything            | Records, contracts, events, projections, workflows and schemas carry explicit versions.        |
| INT-P07 | Idempotent by default         | Commands, event handlers, upload completion and workflow starts tolerate safe retry.           |
| INT-P08 | Facts, not instructions       | Domain events describe completed facts; commands request actions.                              |
| INT-P09 | Minimal payload               | Events and external projections contain only information necessary for their declared purpose. |
| INT-P10 | Traceability                  | Correlation, causation and trace identifiers follow every cross-domain operation.              |
| INT-P11 | Rebuildability                | Projections and analytics can be rebuilt from authoritative sources and event history.         |
| INT-P12 | Failure isolation             | One consumer, provider or derived projection cannot block the source transaction.              |

## 6. Terminology and Core Concepts

| Term                 | Definition                                                                         |
|----------------------|------------------------------------------------------------------------------------|
| Bounded context      | Business boundary with its own language, rules, records and authority.             |
| Aggregate            | Consistency boundary whose changes are validated and versioned together.           |
| Command              | Versioned request asking the authoritative domain to perform an action.            |
| Query                | Permission-safe request for current authoritative or approved projected data.      |
| Domain event         | Immutable statement that a committed business fact occurred.                       |
| Integration event    | Stable reduced event intended for use by another bounded context.                  |
| Projection           | Derived, versioned and permission-scoped representation of source data.            |
| Transactional outbox | Record of committed events written in the same transaction as the business change. |
| Consumer checkpoint  | Durable record proving that a consumer processed an event.                         |
| Dead letter          | Event or command that cannot be safely processed without intervention.             |
| Reconciliation       | Independent comparison between source and target states to identify drift.         |
| Correlation ID       | Identifier connecting all operations belonging to one business journey.            |

## 7. Permanent Architecture Decision

User/External Action

- > Edge authorization
- > Authoritative domain command
- > Local database transaction + outbox
- > Domain event broker
- > Idempotent consumers
- > Projections / Workflow / Notifications / Analytics
- > Reconciliation and operational monitoring

### Architecture decision record

ConstructionOS will use domain-oriented modular architecture with local ACID transactions, versioned synchronous contracts and asynchronous events. It will not use shared writable business tables or distributed transactions across module databases.

**INT-ADR-001 - Domain ownership:** A domain may expose references and projections but cannot delegate its authoritative business validation to an integration consumer.

**Acceptance:** A consumer outage never prevents the source domain from preserving a valid committed transaction.

**INT-ADR-002 - Deployment neutrality:** Logical boundaries must remain valid whether domains are deployed in one modular application or separate services.

**Acceptance:** Extraction of a domain requires contract and deployment changes, not business-record migration between competing owners.

## 8. Bounded Context and Domain Map

| Layer            | Bounded contexts                                                                                                      |
|------------------|-----------------------------------------------------------------------------------------------------------------------|
| Foundation       | Tenant/Group, Company, Identity, User Profile, External Identity, Teams, Authorization, Configuration, Audit, Search  |
| Commercial       | CRM, Units/Sales, Contracts, Procurement, Suppliers, Client Portal, Supplier Portal                                   |
| Project Delivery | Projects, Tasks, Documents, Work Progress, Calendar/Meetings, QA/QC/Handover, HSE, Legal/Compliance, BIM/Digital Twin |

| Layer                      | Bounded contexts                                                                          |
|----------------------------|-------------------------------------------------------------------------------------------|
| Resources                  | Inventory/Warehouses, Assets/Fleet, HR/Payroll/Workforce                                  |
| Financial and Intelligence | Finance, Reporting/Analytics, Notifications/Event Centre, Workflow Engine, Mobile/Offline |

A bounded context is not necessarily one deployed service. It is a mandatory ownership and contract boundary. Shared technical libraries may be used, but shared business persistence is prohibited unless the records belong to a formally defined shared foundation context.

## 9. Source-of-Truth Matrix

| Business domain                    | Authoritative owner | Integration responsibility                      |
|------------------------------------|---------------------|-------------------------------------------------|
| Tenant, group, legal company       | Platform Foundation | All modules reference canonical IDs             |
| Authentication and sessions        | Identity            | All surfaces consume authenticated context      |
| Professional user identity         | User Profile        | Teams/projects display projection               |
| Employment, salary and payroll     | HR/Payroll          | Finance receives posting-ready results          |
| Project and location hierarchy     | Projects            | All project modules reference hierarchy         |
| Unit commercial lifecycle          | Units               | CRM, Finance, QA and Portal consume             |
| Tasks and accountable execution    | Tasks               | Other modules create/link via API               |
| Physical quantities and progress   | Work Progress       | QA/BIM/Reporting consume                        |
| Quality acceptance and handover    | QA/QC               | Work Progress remains separate                  |
| Safety incidents and restrictions  | HSE                 | Projects/HR/BIM consume safe states             |
| Files and controlled revisions     | Documents           | All modules store references                    |
| Formal agreements and obligations  | Contracts           | Finance/Procurement/Legal consume               |
| Purchasing lifecycle               | Procurement         | Inventory/Finance/Portal consume                |
| Stock quantity and movements       | Inventory           | Procurement and Work Progress consume           |
| Asset lifecycle                    | Assets              | Finance/BIM/HSE consume                         |
| Accounting, invoices and payments  | Finance             | Other modules provide source context            |
| Regulatory and legal matters       | Government & Legal  | Projects/HSE/QA consume gates                   |
| Calendar events and meetings       | Calendar/Meetings   | Dates from source modules are projections       |
| Model versions and object identity | BIM/3D              | Operational modules own linked states           |
| Decision orchestration             | Workflow Engine     | Source modules own business consequences        |
| Communication delivery             | Notifications       | Source modules publish facts                    |
| External collaboration             | Portals             | Uses safe projections and immutable submissions |
| Metrics and issued analytics       | Reporting           | Operational modules own facts                   |

## 10. Platform Reference Architecture

### EXPERIENCE

Internal Web | Mobile/PWA | Client Portal | Supplier Portal | BIM Viewer

### EDGE & ACCESS

API Gateway | Authentication | Authorization | Rate Limits | BFFs

### DOMAIN APPLICATIONS

Projects | Units | Tasks | Documents | CRM | Contracts | Finance | Procurement  
Inventory | Assets | HR/Payroll | Work Progress | QA/QC | HSE | Legal | Calendar | BIM

### ORCHESTRATION

Workflow | Notifications | Search | Audit | Configuration | Mobile Sync | Portal Projection

### INTEGRATION BACKBONE

Transactional Outboxes | Broker | Schema Registry | Retry | Dead Letter | Reconciliation

### DATA

Domain Stores | Object Storage | Search Indexes | Audit Store | Warehouse | BIM Artifacts

**INT-ARC-001 - Layering:** Experience applications must use service contracts and cannot connect directly to domain databases.

**Acceptance:** Network and application architecture contains no client-to-database path.

**INT-ARC-002 - Derived services:** Search, Analytics, Portals, Mobile and BIM overlays must consume permission-safe projections rather than become authoritative owners.

**Acceptance:** A projection can be deleted and rebuilt without business-data loss.

## 11. Deployment and Service Decomposition

### Recommended initial deployment

- Modular core application for related lower-volume business domains.
- Strong module boundaries and separate schemas/databases according to risk and scale.
- Independent Documents/file processing service.
- Independent Notifications/Event Centre and broker workers.
- Independent durable Workflow runtime.
- Independent Mobile Sync and upload services.
- Independent Analytics warehouse and query services.
- Independent BIM processing and streaming workers.
- Independent Search indexing infrastructure.

### Extraction criteria

- Independent scale
- Different availability target
- Security isolation
- Special storage technology
- Heavy asynchronous processing
- Independent release cadence
- Operational blast-radius reduction

**INT-DEP-001 - No premature microservices:** A deployment split must be justified by measurable operational need, not by matching one screen to one service.

**Acceptance:** Architecture review documents the reason, ownership and migration path for every extracted service.

## 12. Integration Styles

| Style                                                                                                                                                                                                          | Use when                                                        | Required behavior                                        |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------|----------------------------------------------------------|
| Synchronous command                                                                                                                                                                                            | Caller requires immediate accept/reject/version-conflict result | Authoritative module validates and returns formal result |
| Synchronous query                                                                                                                                                                                              | Current state is required immediately                           | Owner enforces permissions and returns source version    |
| Domain event                                                                                                                                                                                                   | Committed fact should inform independent consumers              | Past-tense immutable event through outbox                |
| Integration event                                                                                                                                                                                              | Cross-domain consumer requires a stable reduced contract        | Minimal and purpose-specific payload                     |
| Projection                                                                                                                                                                                                     | Another surface requires optimized or externally safe data      | Source/version/freshness/permission metadata             |
| Workflow orchestration                                                                                                                                                                                         | Controlled multi-stage decisions are required                   | Source finalization remains authoritative                |
| Analytical ingestion                                                                                                                                                                                           | Historical aggregation and cross-domain analysis                | Warehouse, reconciliation and rebuildability             |
| <b>Selection rule</b><br>Prefer a direct permission-safe query for immediate truth, a command for requested action, an event for an independent reaction, and a projection for optimized repeated consumption. |                                                                 |                                                          |

## 13. Prohibited Integration Patterns

| Prohibited pattern                  | Risk                                    | Required alternative          |
|-------------------------------------|-----------------------------------------|-------------------------------|
| Direct cross-domain database access | Hidden coupling and permission bypass   | API or projection             |
| Shared writable business table      | Competing ownership and unsafe releases | One authoritative owner       |
| Dual write to two databases         | Partial failure and inconsistent state  | Local transaction plus outbox |
| Business logic in broker/ESB        | Unowned rules and opaque behavior       | Owning domain or Workflow     |
| Operational module sends email/push | Duplicate policy and security leakage   | Notification Event Centre     |

| Prohibited pattern                     | Risk                                   | Required alternative                |
|----------------------------------------|----------------------------------------|-------------------------------------|
| Portal writes internal entity directly | External corruption and data exposure  | Immutable external submission       |
| Offline last-write-wins update         | Loss of newer server state             | Versioned offline command           |
| Analytics corrects source value        | Second source of truth                 | Data-quality issue linked to source |
| BIM color changes operational state    | Visual inference becomes business fact | Source module command               |
| File bytes copied between modules      | Duplicate revisions and access drift   | Document/revision reference         |

## 14. Canonical Identity and Cross-Module References

Every cross-module relationship must use immutable platform identifiers. Human-readable codes are display attributes and may change independently.

```
{
  "tenant_id": "uuid",
  "source_module": "procurement",
  "source_type": "purchase_order",
  "source_id": "uuid",
  "source_version": 12,
  "display_code": "PO-2026-0048",
  "deep_link": "/procurement/purchase-orders/uuid"
}
```

**INT-ID-001 - Stable IDs:** IDs are globally unique, immutable, never reused and validated against tenant ownership.

**Acceptance:** Archived records retain their ID and cannot be replaced by a new record using the same ID.

**INT-ID-002 - Reference integrity:** Every polymorphic relationship includes source module, type and ID; high-control relationships include exact source version.

**Acceptance:** Cross-tenant or invalid-type links are rejected server-side.

**INT-ID-003 - Shared party identity:** Client, supplier and external organization records should reference reusable legal/business-party identity where available without merging domain-specific lifecycle data.

**Acceptance:** Procurement supplier status and CRM client status remain separately owned.

## 15. Command Contract Standard

```
{
  "command_id": "uuid",
  "command_name": "inventory.create_receiving_expectation",
  "command_version": 1,
  "tenant_id": "uuid",
  "company_id": "uuid",
  "project_id": "uuid",
  "requested_by": "uuid",
  "source_reference": {"module": "procurement", "type": "purchase_order", "id": "uuid", "version": 9},
  "expected_target_version": 4,
  "idempotency_key": "string",
  "correlation_id": "uuid",
  "payload": {}
}
```

| Result                 | Meaning                                                 |
|------------------------|---------------------------------------------------------|
| Accepted               | Command committed and authoritative result returned     |
| Rejected               | Current business state does not permit requested action |
| ValidationFailed       | Payload or business requirements are incomplete/invalid |
| VersionConflict        | Expected target/source version is stale                 |
| PermissionDenied       | Actor lacks current authority or scope                  |
| TemporarilyUnavailable | Safe retry may be attempted                             |

**INT-CMD-001 - Idempotency:** Every externally retryable or cross-module command requires a caller-supplied idempotency key.

**Acceptance:** Repeated execution returns the original result and creates no duplicate business effect.

**INT-CMD-002 - No assumed success:** The caller must not update its authoritative state until the target returns a committed result or later confirmed event.

**Acceptance:** Timeout is represented as unknown/pending, not success.



## 16. Query Contract Standard

- Authoritative permission evaluation occurs inside the owning domain.
- Responses include record version and source metadata.
- Lists use stable cursor pagination and deterministic sorting.
- Fields are allowlisted by capability and confidentiality.
- Cross-domain aggregation is performed by Reporting, not ad hoc API fan-out from the browser.
- Bulk lookup endpoints accept bounded ID collections to avoid N+1 request patterns.
- Cache headers and freshness semantics are explicit.

**INT-QRY-001 - No inferred access:** Possessing a linked record ID or receiving a summary does not grant access to the authoritative record.

**Acceptance:** Direct open/query is independently authorized by the source module.

**INT-QRY-002 - Projection distinction:** Responses clearly distinguish authoritative live data from cached projection data.

**Acceptance:** UI can display source and last-synchronized state for projections.

## 17. Canonical Event Envelope

```
{
  "event_id": "uuid",
  "event_name": "procurement.purchase_order.issued",
  "event_version": 1,
  "occurred_at": "UTC timestamp",
  "published_at": "UTC timestamp",
  "tenant_id": "uuid", "company_id": "uuid", "project_id": "uuid",
  "actor": {"type": "user", "id": "uuid"},
  "source": {"module": "procurement", "aggregate_type": "purchase_order",
    "aggregate_id": "uuid", "aggregate_version": 9},
  "correlation_id": "uuid", "causation_id": "uuid", "trace_id": "uuid",
  "idempotency_key": "string", "classification": "commercial_restricted",
  "schema_reference": "events/procurement.purchase_order.issued/v1",
  "data": {}
}
```

**INT-EVT-001 - Envelope compliance:** Every production event uses the canonical envelope and registered payload schema.

**Acceptance:** Broker validation rejects malformed or unregistered events from production topics.

**INT-EVT-002 - Minimality:** Events contain only fields necessary for declared consumers and do not include unrestricted document content, credentials or confidential notes.

**Acceptance:** Security review validates classification and payload allowlist.

## 18. Event Naming and Classification

Logical event name format: domain.aggregate.past\_tense\_event. Version is stored separately.

| Class               | Purpose                                         | Example                                   |
|---------------------|-------------------------------------------------|-------------------------------------------|
| Business event      | Committed domain fact                           | contracts.contract.activated              |
| Integration event   | Reduced stable fact for another bounded context | finance.invoice_status_projection_changed |
| System event        | Technical processing state                      | documents.processing_job.failed           |
| Audit event         | Security/governance occurrence                  | security.permission_denied                |
| Notification intent | Communication policy output                     | notifications.intent.created              |
| Analytical event    | Approved fact/delta for warehouse               | analytics.project_progress_fact_changed   |

**INT-EVT-003 - Semantic stability:** An event version cannot change its business meaning or reinterpret an existing field.

**Acceptance:** Breaking meaning creates a new schema version or new event name.

## 19. Event and Contract Catalogue

The Event Centre hosts the central catalogue interface, while each source domain owns and approves its event definitions. The wider contract registry also contains APIs, commands, projections and webhook/adapters where later enabled.

| Catalogue metadata | Requirement                                           |
|--------------------|-------------------------------------------------------|
| Identity           | Name, version, owner, status and business description |

| Catalogue metadata | Requirement                                                 |
|--------------------|-------------------------------------------------------------|
| Source             | Module, aggregate, trigger and ordering key                 |
| Schema             | Payload schema, examples and validation rules               |
| Security           | Classification, retention and permitted consumers           |
| Delivery           | Topic, partition key, replay and idempotency guidance       |
| Lifecycle          | Effective date, deprecation, replacement and end of support |
| Quality            | Contract tests, sample fixtures and production usage        |

**INT-CAT-001 - Registration:** No production contract may be introduced without catalogue ownership, schema and compatibility declaration.

**Acceptance:** CI blocks unregistered event or API deployment.

## 20. Schema Registry and Compatibility

- JSON Schema, Protobuf or equivalent machine-validatable definitions.
- Backward-compatible changes by default.
- Optional fields may be added without changing existing meaning.
- Existing required fields cannot be removed within a version.
- Existing field types and semantics cannot be repurposed.
- Consumers declare supported versions.
- Deprecation includes replacement, migration guide, consumer inventory and end-of-support date.

**INT-SCH-001 - Compatibility check:** Producer CI validates new schemas against configured compatibility mode.

**Acceptance:** Breaking change cannot be published as the same version.

**INT-SCH-002 - Unknown versions:** Consumers route unsupported versions to controlled failure handling rather than silently ignoring required business events.

**Acceptance:** Operations sees an actionable alert and dead-letter record.

## 21. Transactional Outbox Standard

```

BEGIN LOCAL TRANSACTION
  Validate business command
  Update authoritative aggregate and increment version
  Insert audit entry
  Insert outbox event(s)
COMMIT

Outbox relay -> Broker -> Consumer(s)

```

**INT-OUT-001 - Atomic event creation:** Business state and outbox records commit in the same local transaction.

**Acceptance:** Failure before commit produces neither state change nor event; failure after commit remains relayable.

**INT-OUT-002 - Relay safety:** Outbox relay may publish more than once and therefore must preserve stable event IDs.

**Acceptance:** Duplicate broker delivery is handled by consumer idempotency.

**INT-OUT-003 - Retention:** Published outbox rows are retained for an operationally configured period and then archived/cleaned with audit evidence.

**Acceptance:** Retention supports incident investigation and replay policy.

## 22. Event Broker, Partitioning and Ordering

| Concern            | Standard                                                           |
|--------------------|--------------------------------------------------------------------|
| Delivery guarantee | At least once                                                      |
| Ordering           | Per aggregate, not global                                          |
| Partition key      | tenant_id + aggregate_type + aggregate_id                          |
| Topic model        | Domain-oriented topics with controlled system topics               |
| Consumer groups    | One group per independent processing purpose                       |
| Retention          | Configured by event class and replay need                          |
| Security           | Service identity, topic ACL, encryption and payload classification |
| Back pressure      | Bounded concurrency and consumer lag alerts                        |

**INT-BRK-001 - Source transaction isolation:** Broker failure cannot roll back or invalidate an already committed source transaction.

**Acceptance:** Outbox relay resumes when broker availability returns.

## 23. Idempotency, Concurrency and Duplicate Prevention

| Area              | Idempotency / concurrency rule                                |
|-------------------|---------------------------------------------------------------|
| Consumer          | consumer_name + event_id checkpoint                           |
| Entity effect     | consumer + event name + aggregate ID + version where required |
| Command           | Caller idempotency key scoped to tenant and command type      |
| Record update     | Expected version / optimistic concurrency                     |
| File upload       | Upload session + chunk number + checksum                      |
| Workflow start    | Source record + version + trigger + idempotency key           |
| Portal submission | Immutable submission/revision ID                              |
| Mobile command    | Device command ID + expected source version                   |

**INT-CON-001 - Late event handling:** Consumers detect aggregate versions older than their processed checkpoint and apply configured ignore, historical, rebuild or reconciliation behavior.

**Acceptance:** A late event cannot overwrite a newer projection silently.

**INT-CON-002 - No blind overwrite:** Critical source or projection updates must not use last-write-wins without version validation.

**Acceptance:** Conflicts produce explicit state and resolution path.

## 24. Projection Architecture

A projection is a derived read model for a specific purpose. It is replaceable, permission-scoped and never independently editable as business truth.

| Projection type          | Typical consumer         | Required metadata                                      |
|--------------------------|--------------------------|--------------------------------------------------------|
| Calendar projection      | Calendar/Meetings        | Source ID/version/date/deep link                       |
| Portal projection        | External client/supplier | Published version, audience, expiry, withdrawal        |
| Mobile projection        | Registered device        | Workspace, permission snapshot, expiry, source version |
| BIM operational overlay  | Viewer                   | Stable object link, source ID/version/state timestamp  |
| Search document          | Global search            | Permission keys, source version, indexing time         |
| Analytics dimension/fact | Warehouse                | Source lineage, effective date, reconciliation state   |
| Embedded summary         | Project/company pages    | Source owner, freshness and deep link                  |

**INT-PRJ-001 - Rebuildability:** Every projection has a deterministic rebuild path from authoritative sources and registered transformations.

**Acceptance:** Deleting the projection does not delete business records.

**INT-PRJ-002 - Revocation:** Permission or publication revocation removes projection visibility from APIs, search, caches and offline packages according to defined SLA.

**Acceptance:** Revoked access cannot be recovered through stale navigation or enumeration.

## 25. Workflow Integration Contract

| Contract step | Workflow responsibility                                      | Source responsibility                          |
|---------------|--------------------------------------------------------------|------------------------------------------------|
| Submit        | Select workflow and create instance for exact source version | Provide validated record snapshot/reference    |
| Lock          | Request configured lock and record metadata                  | Enforce and confirm field/record lock          |
| Validate      | Invoke registered validation node                            | Return pass/fail/warning with source evidence  |
| Decide        | Record immutable decisions and routing                       | No direct business state change                |
| Finalize      | Request approved transition                                  | Revalidate and commit authoritative transition |

| Contract step | Workflow responsibility          | Source responsibility          |
|---------------|----------------------------------|--------------------------------|
| Complete      | Mark complete after confirmation | Publish resulting domain event |

**INT-WF-001 - Finalization separation:** Approved workflow status is not equal to completed business transition.

**Acceptance:** Workflow remains Source Finalization Pending until source confirmation.

**INT-WF-002 - Material change:** Source modules publish record-version changes that may invalidate configured workflow stages.

**Acceptance:** Invalidated approvals remain visible and cannot silently authorize the changed version.

## 26. Notifications Integration Contract

Operational modules publish facts and source references. The Event Centre evaluates communication policy, resolves authorized recipients, composes safe content and owns delivery state.

- No domain module directly sends in-app, email, push or SMS notifications.
- Recipient access is checked at resolution, composition, external delivery and open time.
- External channels use minimized safe text and authenticated deep links.
- Notification acknowledgement remains separate from formal Workflow approval.
- Deduplication, grouping, quiet hours, digests and retries remain centralized.

**INT-NOT-001 - Source link:** Every business notification retains the authoritative source reference and current permission check.

**Acceptance:** Opening a notification cannot bypass source access.

## 27. Documents Integration Contract

```
{
  "document_id": "uuid",
  "document_revision_id": "uuid",
  "relationship_type": "supporting_evidence",
  "source_record": {"module": "quality", "type": "inspection", "id": "uuid"},
  "visibility_classification": "project_restricted"
}
```

**INT-DOC-001 - No file duplication:** Business modules store Document and exact Revision references instead of copying file bytes or maintaining independent revision history.

**Acceptance:** One underlying file revision can appear in multiple module/folder contexts through references.

**INT-DOC-002 - Exact revision:** Issued decisions, approvals, inspections and contractual records reference the exact controlled revision used at the time.

**Acceptance:** A later superseding revision does not rewrite historical evidence.

**INT-DOC-003 - Access recheck:** Document access is enforced by Documents at view/download time.

**Acceptance:** A visible linked record summary does not grant file access.

## 28. Portal Projection and External Submission Contract

```
Authoritative source
-> External publication policy
-> Permission-safe projection
-> Portal API
-> External user submission (immutable revision)
-> Internal review / Workflow
-> Source module conversion or rejection
-> External-safe outcome projection
```

**INT-POR-001 - Safe publication:** Portal APIs query only published, allowlisted projections and never unrestricted operational entities.

**Acceptance:** External search, counts, URLs and downloads cannot infer hidden records.

**INT-POR-002 - Immutable submission:** Every external quotation, claim, invoice, request, defect, response and acknowledgement is an immutable submission/revision.

**Acceptance:** Internal acceptance creates or updates the source record through a controlled command.

**INT-POR-003 - Relationship access:** An active external account without an active business relationship sees no business records.

**Acceptance:** Access is effective-dated and scoped by company, project, unit, contract or package.

## 29. Mobile and Offline Integration Contract

The mobile layer downloads encrypted, permission-safe workspaces and creates commands rather than editing authoritative database rows.

```
OfflineCommand {
  command_id, device_id, user_id, tenant_id, entity_type, source_id,
  expected_server_version, action, local_timestamp, idempotency_key,
  dependency_ids, evidence_refs, payload
}
```

**INT-MOB-001 - Server revalidation:** Every synchronized command rechecks device trust, current permission, source version and business rules.

**Acceptance:** Local success state is not displayed as synchronized before server confirmation.

**INT-MOB-002 - Conflict safety:** Critical conflicts are represented explicitly and never resolved silently by last-write-wins.

**Acceptance:** User or authorized reviewer can compare local and server versions.

**INT-MOB-003 - Revocation:** Revoked access removes cached records, search entries and file keys when the device reconnects or periodic validation occurs.

**Acceptance:** Pending unauthorized commands are rejected but preserved for audit.

## 30. BIM and Digital Twin Integration Contract

- BIM owns model registry, immutable model versions, federations, derived artifacts and stable object identity.
- Operational links reference stable ConstructionOS object IDs, not renderer node indices.
- Operational colors and overlays are projections of source-module state.
- Selecting a linked object rechecks access to the source record.
- Model revision matching may create exact, high-confidence, ambiguous, split, merge, removed or new outcomes.
- Ambiguous link migration enters a reconciliation queue and cannot be hidden.

**INT-BIM-001 - Stable link:** Cross-module links attach to stable BIM object references and record the model version where created.

**Acceptance:** Renderer optimization or model revision does not silently detach operational history.

**INT-BIM-002 - No reverse ownership:** Changing a visual property or viewer color cannot update progress, QA, HSE, Assets, Units or Finance directly.

**Acceptance:** A user action must invoke an authorized source-module command.

## 31. Reporting and Analytics Integration Contract

```
Operational sources
-> Outbox events / controlled CDC
-> Raw immutable history
-> Transformation + reconciliation
-> Warehouse / materialized aggregates
-> Semantic KPI catalogue
-> Permission-aware query and export
```

**INT-ANA-001 - Operational isolation:** Complex analytical queries run against analytical storage, not source operational databases.

**Acceptance:** Reporting load does not materially affect transactional latency.

**INT-ANA-002 - Permission before aggregation:** Restricted records and fields are removed before totals or averages are calculated.

**Acceptance:** Users cannot infer hidden salary, legal or commercial records from aggregates.

**INT-ANA-003 - Reproducibility:** Facts, dimensions, currency rates, metric versions and snapshots preserve historical effective dates.

**Acceptance:** An issued report can be reproduced from documented versions.

**INT-ANA-004 - No source correction:** Analytics identifies data-quality issues and links to the authoritative record but does not rewrite it.

**Acceptance:** Correction occurs in the source and flows back through normal integration.

## 32. Core Module Event Map

| Producer | Representative events                                                                           | Primary consumers                                                                            |
|----------|-------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------|
| Projects | project.created; project.status_changed; structure.changed; participant_changed; project.closed | Tasks, Documents, Finance, Work Progress, QA, HSE, BIM, Calendar, Reporting, Mobile, Portals |

| Producer          | Representative events                                                                                     | Primary consumers                                                              |
|-------------------|-----------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------|
| Units             | unit.created; unit.status_changed; unit.reserved; buyer_assigned; handover_status_changed                 | CRM, Contracts, Finance, QA, Client Portal, BIM, Reporting                     |
| Documents         | document.created; revision.uploaded; revision.published; revision.superseded; access.changed              | All linked modules, Workflow, Notifications, Portals, Mobile, BIM              |
| Tasks             | task.created; task.assigned; task.status_changed; task.overdue; task.completed                            | Calendar, Notifications, Projects, QA, HSE, Mobile, Reporting                  |
| Contracts         | contract.signed; contract.activated; contract.amended; obligation.overdue; contract.terminated            | Finance, Procurement, Projects, Units, Legal, QA, Portals, Reporting, Workflow |
| Procurement       | requisition.submitted; rfq.issued; quotation.received; purchase_order.issued; delivery.delayed            | Workflow, Portal, Inventory, Assets, Contracts, Finance, QA, Reporting         |
| Inventory         | receipt.posted; stock_movement.posted; stock_below_minimum; item.quarantined/released                     | Procurement, Work Progress, QA, Assets, Finance, Mobile, Reporting             |
| Assets            | asset.created; asset.assigned; inspection_due; maintenance_completed; asset.disposed                      | Projects, Calendar, HSE, QA, Finance, BIM, Mobile, Reporting                   |
| Work Progress     | entry.submitted/accepted; quantity.changed; scope_ready_for_inspection; rework_required                   | Projects, QA, HSE, Tasks, HR, Finance, BIM, Reporting                          |
| QA/QC             | inspection.passed/failed; ncr.issued/closed; defect.created/closed; handover.readiness_changed            | Work Progress, Tasks, Documents, Procurement, Assets, Units, HSE, BIM, Portals |
| HSE               | hazard.reported; incident.classified; stop_work.issued/released; eligibility.changed                      | Projects, Tasks, HR, Assets, QA, BIM, Mobile, Portals                          |
| HR/Payroll        | employment_activated; leave.approved; timesheet.locked; payroll_run.finalized; employee.offboarded        | Teams, Projects, Work Progress, HSE, Finance, Calendar, Mobile, Workflow       |
| Finance           | budget.approved; commitment.recorded; invoice.posted; payment.posted; period.closed; transaction.reversed | Projects, Contracts, Procurement, Units, HR, Assets, Portals, Reporting        |
| Legal             | permit.granted/suspended; obligation.overdue; notice.issued; restriction.activated/released               | Projects, Contracts, HSE, QA, Documents, Finance, Calendar, Workflow           |
| Calendar/Meetings | event.rescheduled; minutes.issued; decision.recorded; action_item.created                                 | Notifications, Tasks, Projects, Portals, Reporting                             |
| BIM               | model_version.published; federation.published; object_mapping.changed; reconciliation_required            | Projects, Work Progress, QA, HSE, Assets, Units, Mobile, Portals               |
| Workflow          | instance.started; decision.recorded; instance.approved/rejected; source_finalization_requested; completed | Source module, Notifications, Audit, Reporting, Mobile, Portals                |
| Portals           | submission.received; quotation/claim/invoice.submitted; acknowledgement.recorded; access.revoked          | Source modules, Workflow, Notifications, Audit                                 |
| Mobile            | command.received/accepted/rejected; conflict.created; file.uploaded; device.revoked                       | Source modules, Audit, Notifications, Operations                               |

### 33. Procurement-to-Pay Reference Flow

Demand -> Requisition -> Workflow Approval -> RFQ -> Supplier Portal  
-> Immutable Quotation -> Evaluation/Award -> Workflow Approval -> Purchase Order  
-> Inventory Receiving Expectation -> Delivery -> Receipt + QA Result  
-> Supplier Invoice -> Finance Posting -> Payment Workflow -> Payment Posted  
-> Portal Status + Notifications + Reporting

| Step                   | Authoritative owner             | Integration result                                     |
|------------------------|---------------------------------|--------------------------------------------------------|
| Demand and requisition | Procurement                     | Workflow trigger and budget query                      |
| RFQ and quotation      | Procurement / Portal submission | Supplier cannot view competitors                       |
| Award and PO           | Procurement                     | Approved PO event creates receiving/commitment context |
| Receipt                | Inventory                       | Exact PO line references and quantity outcomes         |
| Quality inspection     | QA/QC                           | Accepted, rejected or quarantined result               |
| Invoice and payment    | Finance                         | Three-way references and accounting truth              |
| External visibility    | Supplier Portal                 | Permission-safe status projection                      |

**INT-FLOW-001 - No ownership crossover:** Procurement does not post stock or accounting; Inventory does not change PO commercial terms; Finance does not rewrite receiving quantities.

**Acceptance:** End-to-end test verifies references and authoritative owners at every step.

## 34. Progress-to-Quality-to-Handover Reference Flow

Work Progress submits completed scope  
 -> QA/QC inspection readiness  
 -> Pass: accepted quality state / Fail: NCR or Defect + corrective Task  
 -> Reinspection  
 -> Handover Readiness recalculated from mandatory sources  
 -> Workflow decision  
 -> QA/QC records accepted Handover  
 -> Client Portal receives approved projection

**INT-FLOW-002 - Progress-quality separation:** Physical completion cannot automatically create quality acceptance or handover readiness.

**Acceptance:** A 100% progress record may remain unaccepted when inspection or mandatory evidence is incomplete.

**INT-FLOW-003 - Readiness source links:** Every handover blocker links to its authoritative source record and version.

**Acceptance:** Readiness score cannot override a mandatory blocker.

## 35. Contract-to-Finance Reference Flow

CRM / Procurement / Project requests agreement  
 -> Contracts prepares controlled agreement + Document revisions  
 -> Workflow legal/commercial approval  
 -> Contract activated  
 -> Finance creates authorized structures and commitments  
 -> Invoices/payments reference Contract ID and effective version  
 -> Reporting analyzes value, obligations and performance

**INT-FLOW-004 - Effective contract version:** Financial and procurement transactions reference the effective contract/amendment version used for the transaction.

**Acceptance:** Later amendments do not rewrite historical posting context.

**INT-FLOW-005 - Contract activation:** Finance structures are created only after a confirmed contract activation or explicit approved exception.

**Acceptance:** Duplicate activation event does not create duplicate financial structures.

## 36. HR-to-Payroll-to-Finance Reference Flow

Employment + compensation + schedules  
 -> Attendance / Timesheets  
 -> Payroll calculation and validation  
 -> Workflow approval  
 -> Payroll finalized  
 -> Posting-ready summary to Finance  
 -> Finance journals/payments  
 -> Posting references returned  
 -> Payslip stored in Documents + Notification

**INT-FLOW-006 - Confidentiality boundary:** Project and Reporting integrations cannot infer individual salary from labour-cost or payroll projections without explicit permission.

**Acceptance:** Unauthorized users receive approved aggregate/cost projections only.

**INT-FLOW-007 - Payroll posting reconciliation:** Every finalized payroll run reconciles to Finance posting and payment status.

**Acceptance:** Missing or mismatched posting enters a visible reconciliation queue.

## 37. Security and Authorization Architecture

| Authorization point        | Required check                                                |
|----------------------------|---------------------------------------------------------------|
| API Gateway                | Authentication, tenant, rate limit and request policy         |
| Authoritative domain       | Capability, company, project, record, field and business rule |
| Event recipient resolution | Current access and relationship, not event payload alone      |
| Projection generation      | Allowlisted fields and target audience                        |
| Portal publication         | External relationship, effective scope and publication policy |
| Mobile download/sync       | Device trust, workspace scope, expiry and current access      |



| Authorization point | Required check                                         |
|---------------------|--------------------------------------------------------|
| Documents           | Document/revision classification and link-target scope |
| BIM object link     | Viewer and linked source-record permission             |
| Analytics           | Row/field/metric restriction before aggregation        |
| Export              | Export capability, watermark and audit                 |

**INT-SEC-001 - No transitive permission:** Access to a projection, notification, task, model object or linked summary does not grant access to the target source record.

**Acceptance:** Target module independently authorizes every open and command.

**INT-SEC-002 - Service identity:** Every producer and consumer authenticates with least-privilege service credentials and topic/API scopes.

**Acceptance:** Compromised consumer cannot publish arbitrary source-domain events.

**INT-SEC-003 - Sensitive logging:** Logs, traces, dead letters and metrics must not expose confidential payloads beyond authorized operational access.

**Acceptance:** Security review confirms redaction and retention.

## 38. Audit and Traceability Architecture

| Audit class                                                                                                                                                                               | Owner                         | Examples                                                     |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------|--------------------------------------------------------------|
| Domain audit                                                                                                                                                                              | Authoritative module          | Business before/after, actor, reason, source command         |
| Integration audit                                                                                                                                                                         | Integration platform/consumer | Event published/received, processing result, retry           |
| Workflow audit                                                                                                                                                                            | Workflow Engine               | Selection, participants, decisions, finalization             |
| Notification audit                                                                                                                                                                        | Event Centre                  | Recipient decision, delivery, read, acknowledgement          |
| Security audit                                                                                                                                                                            | Security/Audit service        | Permission denial, sensitive read, export, override          |
| External audit                                                                                                                                                                            | Portal                        | Invitation, access, download, submission and acknowledgement |
| Device audit                                                                                                                                                                              | Mobile                        | Workspace, offline command, conflict, device revocation      |
| trace_id -> correlation_id -> command_id -> source aggregate/version<br>-> event_id(s) -> workflow_instance_id -> notification_id<br>-> projection_version -> consumer processing records |                               |                                                              |

**INT-AUD-001 - End-to-end trace:** Support can trace one business action across all participating services using correlation and causation metadata.

**Acceptance:** Operational console resolves a correlation ID to all related technical and business references.

## 39. Failure, Retry and Dead-Letter Handling

| Failure                                   | Retry?             | Required behavior                               |
|-------------------------------------------|--------------------|-------------------------------------------------|
| Network timeout / temporary service error | Yes                | Exponential backoff, jitter, bounded attempts   |
| Rate limit                                | Yes                | Respect retry-after and consumer throttling     |
| Permission denied                         | No                 | Record permanent rejection and security context |
| Business validation failed                | No automatic retry | Return actionable source result                 |
| Version conflict                          | No blind retry     | Refresh source and resolve conflict             |
| Unknown schema version                    | No                 | Dead letter and alert owner                     |
| Dependency temporarily missing            | Conditional        | Retry or wait using configured policy           |
| Duplicate event/command                   | No new effect      | Return/checkpoint original result               |

**INT-DLQ-001 - Dead-letter record:** Dead letters preserve original envelope, consumer, failure class, attempts, source reference, replay eligibility and resolution history.

**Acceptance:** Operator can determine whether replay, repair or discard is safe.

**INT-DLQ-002 - Replay safety:** Replay rechecks current schema, permission, source state and obsolescence before business effects occur.

**Acceptance:** Replaying a stale event cannot recreate a cancelled or superseded action silently.



## 40. Reconciliation Architecture

Events provide timely propagation; reconciliation provides correctness assurance. Critical integration pairs require scheduled and on-demand comparison.

| Source                 | Target / comparison                                      |
|------------------------|----------------------------------------------------------|
| Purchase Orders        | Finance commitments and Inventory receiving expectations |
| Deliveries             | Inventory receipts and QA outcomes                       |
| Supplier invoices      | Finance payable records                                  |
| Payroll runs           | Finance journals and payments                            |
| Work Progress          | QA accepted scope and handover readiness                 |
| Asset purchases        | Activated Asset records and Finance asset references     |
| Contracts              | Invoices, payments, obligations and amendments           |
| Portal publications    | Published source versions and current relationships      |
| BIM links              | Stable object versions and operational targets           |
| Mobile commands        | Source records and final sync result                     |
| Analytics              | Operational totals and source counts                     |
| Outcome                | Meaning                                                  |
| Matched                | Source and target agree                                  |
| Missing Target         | Expected downstream record/projection absent             |
| Missing Source         | Orphan target/projection exists                          |
| Version Mismatch       | Target references stale or unexpected version            |
| Value Mismatch         | Critical quantities/amounts do not reconcile             |
| Permission Mismatch    | Projection visibility differs from current authority     |
| Stale Projection       | Refresh SLA exceeded                                     |
| Manual Review Required | Automated correction is unsafe                           |

**INT-REC-001 - No silent auto-fix:** Reconciliation may safely rebuild derived projections but cannot automatically rewrite authoritative business records without a source-domain command.

**Acceptance:** Every correction path is owned and audited.

## 41. Observability and Operational Support

| Metric / signal                       | Purpose                                                     |
|---------------------------------------|-------------------------------------------------------------|
| Outbox backlog age                    | Detect unpublished committed facts                          |
| Broker consumer lag                   | Detect delayed downstream processing                        |
| Processing success/failure rate       | Measure consumer health                                     |
| Duplicate suppression                 | Detect unstable producers or retries                        |
| Dead-letter count/age                 | Surface unresolved integration failures                     |
| Projection freshness                  | Expose stale Calendar, Portal, Mobile, BIM and Search state |
| Workflow source-finalization failures | Detect approved but unapplied decisions                     |
| Reconciliation mismatches             | Detect source-target drift                                  |
| Mobile pending queue age              | Detect field data not reaching source modules               |
| Analytics ingestion delay             | Expose reporting staleness                                  |

**INT-OBS-001 - Distributed tracing:** Every synchronous and asynchronous hop propagates trace and correlation context.

**Acceptance:** A single trace view shows API, database, outbox, broker and consumer timing where supported.

**INT-OBS-002 - Runbooks:** Every critical consumer, reconciliation job and dead-letter queue has an operational owner and recovery runbook.

**Acceptance:** On-call personnel can distinguish safe retry from business intervention.

## 42. API Gateway, BFF and Edge Architecture

- One external entry layer for authentication, tenant context, rate limiting and request tracing.
- Separate Backend-for-Frontend contracts may exist for internal web, mobile and portals.
- BFFs compose permitted views but do not own business records or bypass source authorization.
- Internal service APIs remain inaccessible from public networks unless explicitly exposed through a controlled adapter.
- Large files use signed upload/download sessions through Documents or Mobile File services.
- Real-time updates use authenticated WebSocket/SSE channels carrying minimal event references.

**INT-EDGE-001 - Active context:** Every request resolves tenant, active company and user/external organization context before reaching a domain.

**Acceptance:** Cross-company action without explicit capability and context is rejected.

**INT-EDGE-002 - Rate limits:** Rate limits differ by internal user, external portal, device sync, file upload and service-to-service traffic.

**Acceptance:** Abuse or tender/upload spikes cannot degrade core transactional APIs uncontrolled.

## 43. Data and Storage Architecture

| Storage                  | Purpose                                     | Ownership rule                                     |
|--------------------------|---------------------------------------------|----------------------------------------------------|
| Domain relational stores | Authoritative transactional records         | One bounded context writes each business aggregate |
| Object storage           | Documents, media and BIM artifacts          | Access through owning service and signed policies  |
| Event broker             | Transient/durable event transport           | Not authoritative business storage                 |
| Audit store              | Append-only security/integration evidence   | Immutable and retention governed                   |
| Search indexes           | Permission-aware discovery                  | Derived and rebuildable                            |
| Reporting warehouse      | Historical facts, dimensions and aggregates | Derived from sources; no source writes             |
| Projection stores        | Portal, Mobile, Calendar, BIM overlays      | Purpose-specific and replaceable                   |
| Cache                    | Performance acceleration                    | Never sole store for committed business state      |

**INT-DATA-001 - Tenant ownership:** Every persisted record, event, projection and file metadata entry includes tenant ownership and validates foreign references within scope.

**Acceptance:** Cross-tenant foreign key or polymorphic link tests fail closed.

**INT-DATA-002 - Money and units:** Cross-module financial and quantity contracts include currency/unit explicitly and use decimal-safe representations.

**Acceptance:** No floating-point monetary integration field is accepted.

## 44. Platform Configuration and Administration

- Enable/disable events and consumers by tenant/company/module.
- Configure projection publication and freshness SLAs.
- Configure retry, backoff, dead-letter and reconciliation schedules.
- Manage contract ownership, versions, deprecation and consumer inventory.
- Configure security classifications and retention.
- Configure broker topics, quotas and service identities through controlled operations.
- View integration health, outbox backlog, lag, failures and mismatches.
- Perform authorized replay, rebuild and reconciliation with audit history.

**INT-ADM-001 - Administrative separation:** Integration administration does not automatically grant access to confidential business payloads or source records.

**Acceptance:** Operations views use redacted metadata unless separately authorized.

**INT-ADM-002 - Change governance:** Production contract, routing and replay configuration is versioned and subject to controlled change management.

**Acceptance:** Configuration history identifies actor, reason, effective time and rollback version.

## 45. Non-Functional Requirements

| Area         | Requirement                                                                    |
|--------------|--------------------------------------------------------------------------------|
| Availability | Source domains remain available when non-critical consumers or providers fail. |

| Area              | Requirement                                                                                                     |
|-------------------|-----------------------------------------------------------------------------------------------------------------|
| Durability        | Committed outbox events, workflow state and audit records survive service restart and deployment.               |
| Performance       | Common synchronous commands/queries meet module targets; event propagation target is configured by criticality. |
| Scalability       | Broker, consumers and projection workers scale horizontally by tenant/aggregate partitions.                     |
| Security          | Encryption in transit/at rest, least privilege, secret rotation and payload minimization.                       |
| Data residency    | Storage and processing region policy remains configurable for regulated deployments.                            |
| Accessibility     | Integration failure and stale-state UI is understandable without color-only indicators.                         |
| Maintainability   | Shared SDKs, generated contracts, fixtures and observability reduce per-module custom code.                     |
| Disaster recovery | Backups, broker retention, rebuild and replay procedures meet defined RPO/RTO.                                  |
| Privacy           | Events and projections apply purpose limitation, minimization, retention and legal hold.                        |

**INT-NFR-001 - Failure isolation:** One failed consumer or projection cannot fail the complete dashboard, source transaction or unrelated consumer group.

**Acceptance:** Chaos test demonstrates isolated consumer failure and recovery.

**INT-NFR-002 - Back pressure:** High-volume producers and large tenant workloads are controlled by quotas, partitions and bounded concurrency.

**Acceptance:** Broker lag alerts before user-visible SLA breach.

## 46. Testing and Certification Strategy

### Contract tests

- Producer schema and envelope validation.
- Consumer supported-version validation.
- Duplicate and late-event handling.
- Authorization and classification tests.
- Idempotent command and consumer behavior.
- Projection source/version/freshness validation.

### Mandatory end-to-end scenarios

1. Procurement requisition through supplier payment.
2. Work Progress through inspection, defect correction and handover.
3. Employee timesheet through payroll and Finance posting.
4. Client defect submission through QA/QC closure projection.
5. Supplier quotation through approved Purchase Order.
6. Offline HSE incident through synchronized source investigation.
7. BIM object link across a model revision and ambiguous mapping.
8. Contract amendment invalidating an active approval workflow.
9. Permission revocation removing Portal, Mobile, Search and Document access.
10. Dead-letter replay after the source state has changed.

**INT-TST-001 - Certification gate:** A module cannot activate a production integration until it passes standard contract, security, duplicate, recovery and reconciliation tests.

**Acceptance:** Certification evidence is stored in the integration registry.

## 47. Acceptance Criteria

**AC-001 - Acceptance criterion:** Every governed business entity has one documented authoritative owner.

**AC-002 - Acceptance criterion:** No production module directly reads or writes another module database.

**AC-003 - Acceptance criterion:** Every cross-domain write uses a versioned command or source-owned API.

**AC-004 - Acceptance criterion:** Every committed cross-domain fact is emitted through transactional outbox or approved controlled CDC.

**AC-005 - Acceptance criterion:** Every consumer is idempotent and passes duplicate-delivery testing.

- AC-006 - Acceptance criterion:** Every production API, command, event and projection is registered and versioned.
- AC-007 - Acceptance criterion:** Every projection identifies source module, source record, source version and freshness.
- AC-008 - Acceptance criterion:** Notifications are delivered only through the Notifications & Event Centre.
- AC-009 - Acceptance criterion:** Workflow completion requires source-module finalization confirmation.
- AC-010 - Acceptance criterion:** Portal users cannot directly modify authoritative internal entities.
- AC-011 - Acceptance criterion:** Offline clients submit versioned commands and do not use blind last-write-wins updates.
- AC-012 - Acceptance criterion:** Files are linked by Document/Revision reference without duplicate version histories.
- AC-013 - Acceptance criterion:** BIM operational links use stable object identity and source permission checks.
- AC-014 - Acceptance criterion:** Analytics can be rebuilt and reconciled to operational sources.
- AC-015 - Acceptance criterion:** Permission revocation propagates to APIs, projections, caches, search and offline packages.
- AC-016 - Acceptance criterion:** Dead-letter, replay, rebuild and reconciliation actions are audited.
- AC-017 - Acceptance criterion:** One correlation ID can trace a reference flow across all participating services.
- AC-018 - Acceptance criterion:** Critical cross-module pairs have automated reconciliation and visible exception queues.
- AC-019 - Acceptance criterion:** Cross-tenant, cross-company, cross-project and external-isolation test suites pass.
- AC-020 - Acceptance criterion:** Operational dashboards, alerts and runbooks are production-ready before rollout.

## 48. Implementation Phases and Sequencing

| Phase                                                                                                                                                                                                                                                                                     | Scope                                                                                                                            |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------|
| Phase 1 - Integration foundation                                                                                                                                                                                                                                                          | Canonical IDs and context; authorization contract; command/event envelopes; outbox SDK; broker; schema registry; trace standard. |
| Phase 2 - Reference vertical flow                                                                                                                                                                                                                                                         | Contract activation -> Finance structure -> invoice -> approval -> payment -> projection -> analytics -> notification.           |
| Phase 3 - Project delivery                                                                                                                                                                                                                                                                | Projects, Tasks, Documents, Work Progress, QA/QC, HSE, Inventory and Assets.                                                     |
| Phase 4 - Commercial and workforce                                                                                                                                                                                                                                                        | CRM, Units, Contracts, Procurement, Suppliers, Finance, HR/Payroll and Legal.                                                    |
| Phase 5 - Orchestration                                                                                                                                                                                                                                                                   | Workflow, Notifications, Calendar projections, Search and federated audit.                                                       |
| Phase 6 - External and field                                                                                                                                                                                                                                                              | Client/Supplier Portals, Mobile/Offline and BIM operational overlays.                                                            |
| Phase 7 - Intelligence                                                                                                                                                                                                                                                                    | Warehouse, semantic layer, data quality, reconciliation and executive analytics.                                                 |
| <b>Recommended reference implementation</b><br>Implement one complete vertical business journey first. Use its command wrapper, authorization, transaction, audit, outbox, consumer, projection, notification and reconciliation behavior as the certified pattern for all other modules. |                                                                                                                                  |

## 49. Migration and Adoption Strategy

1. Inventory existing direct database queries, dual writes, copied fields, direct notifications and module-specific approval paths.
  2. Assign authoritative ownership to every shared entity and calculated value.
  3. Introduce canonical identifiers and cross-module reference objects.
  4. Add shared command, query, outbox, event and trace libraries.
  5. Create permission-safe projections alongside existing reads.
  6. Run new consumers in shadow mode and compare results through reconciliation.
  7. Migrate one vertical flow at a time behind automated tests and feature flags.
  8. Disable legacy coupling only after parity, security and recovery acceptance.
  9. Monitor contract usage and remove deprecated contracts after all consumers migrate.
- INT-MIG-001 - Historical preservation:** Migration does not rewrite historical source ownership, record versions, issued documents or audit events.
- Acceptance:** Legacy references remain resolvable and clearly identified.

## 50. Risks and Open Decisions

| Risk / decision                              | Required mitigation or decision                                                                    |
|----------------------------------------------|----------------------------------------------------------------------------------------------------|
| Over-engineering with excessive services     | Use logical contexts first; extract only with scale/security/availability justification.           |
| Inconsistent IDs and polymorphic links       | Canonical ID/reference standard and tenant ownership validation.                                   |
| Event payload growth and data leakage        | Schema ownership, classification and payload minimization review.                                  |
| Consumer drift after schema change           | Registry compatibility checks, consumer inventory and deprecation monitoring.                      |
| Unbounded eventual consistency               | Projection SLAs, visible freshness, reconciliation and synchronous verification for final actions. |
| Replay creates obsolete effects              | Current-state validation and event-specific replay policy.                                         |
| Analytics or search infer restricted records | Permission projections before indexing/aggregation and inference tests.                            |
| Shared foundation becomes a monolith         | Narrow foundation ownership and no business-domain rules in shared services.                       |
| Broker/provider selection                    | Open decision based on hosting, scale, operations skills and residency.                            |
| Database-per-domain timing                   | Open decision by deployment stage; logical ownership is mandatory regardless.                      |
| Multi-region topology                        | Decide residency, replication and failover before regulated-market rollout.                        |
| Event retention                              | Set by event class, audit need, rebuild strategy and cost.                                         |

## 51. Definition of Done

- All Must requirements and acceptance criteria are implemented and tested.
- Complete source-of-truth and bounded-context registry is approved.
- Canonical command, query, event, projection and trace SDKs are available.
- Transactional outbox and idempotent consumer patterns are production certified.
- API and event schemas are registered with owners and compatibility policy.
- At least one complete reference flow runs in production using the new architecture.
- Security review confirms tenant/company/project/record isolation at every boundary.
- Dead-letter, replay, reconciliation and projection rebuild operations are tested.
- Operational dashboards, alerts, ownership and runbooks are approved by SRE.
- No unresolved critical direct database coupling or dual-write issue remains in migrated flows.
- Document, Mobile, Portal, BIM, Workflow, Notification and Analytics contracts conform to this PRD.
- Architecture governance process is established for future modules and integrations.

## 52. Integration Registry and Required Deliverables

| Deliverable              | Required content                                                                 |
|--------------------------|----------------------------------------------------------------------------------|
| Source-of-Truth Registry | Entity, authoritative domain, lifecycle, confidentiality and allowed projections |
| API Contract Registry    | Endpoint, version, owner, auth, schemas, consumers and deprecation               |
| Event Catalogue          | Event name/version, schema, source, consumers, classification and ordering       |
| Command Catalogue        | Command name/version, source, target, idempotency and result model               |
| Projection Registry      | Purpose, source, target, fields, permissions, freshness and rebuild method       |
| Integration Matrix       | Producer, consumer, contract, direction, criticality and reconciliation          |
| Trace Map                | Correlation and causation behavior across reference flows                        |
| Operational Runbooks     | Failure, retry, dead letter, replay, rebuild and reconciliation procedures       |
| Certification Pack       | Contract, security, concurrency, recovery and end-to-end test evidence           |
| Deprecation Register     | Legacy integration, replacement, consumer migration and retirement date          |